

QML Explainable AI Dashboard: Early Detection in ICU

1. Concept & Visual Design (Neobrutalism UI)

ICU me doctors ke paas har waveform ko manually analyze karne ka time nahi hota. Isliye, yeh dashboard ek pure dark theme par based hai (Neobrutalism style), jisme sharp borders aur high-contrast neon colors hain. Isse doctor ki aankh seedha 'minor change' par jayegi bina kisi screen clutter ke.

2. The Scenario: ICU Bed No. 4

- Patient: 55-year-old male, kal intestinal surgery hui hai.
- Current Vitals (Normal Monitor): Heart Rate 85 bpm, BP 118/75, SpO2 97%.
- Initial View: Sab normal hai, patient stable lag raha hai. Par QML surface ke neeche chal rahi tabahi ko dekh raha hai.

3. Widget 1: The "Waveform Ghosting" (Seeing the Invisible Delay)

- Visual UI: Ek grey color ki 'Ghost Wave' (patient ki 12 ghante pehle ki normal wave) ke upar ek bright neon red wave (live data) chal rahi hai.
- The Minor Change: Neon red wave grey wave ke muqable sirf 4 milliseconds (ms) right side shift ho gayi hai. Yellow warning arrow is delay ko point karta hai.
- Medical Implication: Ye diastolic notch ka delay hai. Ye batata ki vascular resistance kam ho rahi hai (nasein dheeli pad rahi hain), jo Sepsis ka 24 ghante pehla sign hai.
- The Mathematics/Rules Used: Quantum Phase Estimation (QPE) aur Hilbert-Huang Transform. Classical Fourier transform is time-localization ko kho deta hai. QML QPE ka use karke phase space me 4ms ke minor shift ko mathematically magnify karta hai, jo classical sensors ke liye 'noise' tha.

4. Widget 2: The Entanglement Network (Organ Sync Map)

- Visual UI: Screen par 3 glowing circles hain (Heart, Lungs, Kidneys) jo thick green neon lines se jude hain.
- The Minor Change: Heart aur Lungs ke beech ki green line orange ho jati hai aur patli hone lagti hai. Text floats: "Cardio-Respiratory Decoupling: 14% Loss in Sync."
- Medical Implication: Saans lene ki rhythm aur heart beats ka aapas me jo natural sync hota hai, wo toot raha hai. Ye body ke internal defense system ke fail hone ka sabse early sign hai.
- The Mathematics/Rules Used: Quantum Entanglement Entropy (Von Neumann Entropy). QML in organs ke time-series data ko quantum state me entangle karta hai. Jab physiological sync toot-ta hai, toh entanglement entropy mathematically drop hoti hai. QML is correlation loss ko detect karke alert karta hai.

5. Widget 3: The HRV (Heart Rate Variability) Micro-Trend

- Visual UI: Ek bar chart jisme bars ki height lagatar ek jaisi (flat) hoti ja rahi hai.
- The Minor Change: Alert dikhata hai: "HRV dropped from 45ms to 28ms in the last 4 hours."
- Medical Implication: Heart ek rigid machine ki tarah exact interval par dhadak raha hai. Uski natural flexibility (variability) khatam ho gayi hai, jo extreme physiological stress (shock) ko darshata hai.
- The Mathematics/Rules Used: Non-linear Dynamics & Lyapunov Exponents in Hilbert Space. QML patient ke RR-intervals ka largest Lyapunov exponent calculate karta hai. Jab system chaos se periodic state (rigidity) me jata hai, exponent ki value zero ki taraf shift hoti hai. Ye pure mathematics autonomic nervous system ki thakawat ko measure karta hai.

6. The Final Predictive Alert & Doctor's Action

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- Final Alert on Screen (High-Contrast Neon Red):

- * PREDICTIVE ALERT: BED 4

- * Condition: 88% Probability of Septic Shock.

- * Estimated Time to Crash: 14 to 18 Hours.

- * Visual Evidence: Dicrotic notch delayed by 4ms (Vasodilation) + 14% Cardio-Respiratory Decoupling.

- The Result: Doctor ko ab AI par blindly trust nahi karna padega kyunki usne math ko visual proof (Ghost wave, Decoupling) me dekh liya hai. Patient ke crash hone se 14 ghante pehle hi Lactate test aur Antibiotics shuru ho jayenge, aur patient bach jayega.